Taxonomy for Human vs. AI understanding in Mathematics

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Three tiers of AI contribution

- complete human understanding (Almost as before AI)
  - AI as a writing assistant (formatting, language, search, coding)
  - In dialogue the AI provided small pieces of knowledge, as if one read them in a paper.
- partial human understanding, AI-conceived discovery
  - verified by humans or Lean
  - some significant part was autonomously generated by AI with minimal human oversight
  - Proof is understood by humans, but its consequences were not explored
- no appreciable understanding
  - LLM generated
  - no verification (neither human nor Lean)
  - not structured, digested, understood by humans
  - not welcome on arXiv

My questions:
- Does the researcher understand all parts of the paper?
- Was the discovery guided by the human or AI?

The point is to be able to talk about the depth of human understanding and the extent of AI contribution.

Enforcibility is not a primary concern for now.

Class 0
Zero AI

No use of LLMs. Exceptions are strictly limited to

- spell checkers (Grammar, orthography, prose rewrite)
- code linters
- formatting / typeset (LaTeX)
- Searching for literatur refereces.
  Only the existance not the content of a paper.

Guideline:
If the LLM did something that a conventional spell checker, an older machine learning tool, LaTeX-package description, or Google Search could have done, then it is still `class 0`

`Text, Lit` but restricted

Class 1
Human-conceived discovery, AI as research assistant

LLMs can be used for

- writing exposition, introduction, known general theory
- all language improvement
- presentation, layout, diagrams, tables
- clarity and elegance of formulas
- cross-references
- Literature work: search, summary, citations, bibliography
- coding, simulations
- data work: collection, labelling, plots
- translation (if verified)
- **deterministic** calculations

`Text, Lit, Code, Data, Idea` unrestricted

`Class 1`
Human-conceived discovery, AI as research assistant

LLMs can also be used for

- `Check`: verifying an argument
- `Proof`: **localized** mathematical ideas
- `Result`: Ideas for generalizing or dropping an assumption
- `Method`: Talking about the big picture proof strategy

`Check, Proof, Result, Method` restricted

Class 1
Human-conceived discovery, AI as research assistant

- Progress in understanding is completely determined by the human.
- Only **localized** additional ideas are allowed to be provided by AI.
- Human always thinks ahead and anticipates the LLM's answer
- All **AI "contributions"** are already part of **math literature**
- **The entire content was conveived by a human first.**
  Every piece of information was cited from another paper or discovered by the authors. Nothing originates with the AI.

`Text`, `Lit`, `Code`, `Data`, `Idea` unrestricted,
`Check`, `Proof`, `Result`, `Method` restricted

Class 2
AI-prompted discovery with human understanding

At least **one significant mathematical idea, claim, or proof originates directly from AI**, without finding an exact reference in the math literature of this idea. The researcher must understand the idea/claim/proof and engage with and explore this idea in a more general context than the proof the AI provided.

`Check, Result` unrestricted,
`Proof, Method` restricted

Class 2
AI-prompted discovery with human understanding

- AI-Check generates novel arguments
- Improvements to the Result (statement, assumptions) can originate from AI.

Check, Result unrestricted,

- Proof, proofstrategy, structure of proof, connection between arguments, connection to other papers can **locally** originate from AI
- **Big picture strategy must be human.**

Proof, Method restricted

Class 2
AI-prompted discovery with human understanding

- All AI contributions were tightly controlled by the human researchers.
- The researcher is the driver: **human activity**
- Only small improvements, local changes, thought-provoking ideas, connections to other fields, forgotten consequences originate from AI.
- The researcher made an effort to understand every part of the material before moving on.
- Little by little AI helped the human to make progress.
- Never did the AI production outpace the human understanding.

`Proof, Method` restricted

Class 3
Partial AI Generation

Entire claims, chunks of proofs, or whole sections were written directly by an AI **without prior conceptual understanding by the human authors.**

Even if the human author thoroughly checked, verified, and fully understood the AI's generated work, the fact that a **whole cohesive section of mathematical understanding originated from the AI automatically triggers class 3.**

`Proof, Method` not mostly autonomous

Class 3
Partial AI Generation

- The human is **not obligated to engage, understand, and work with the AI output in a deeper way.** The authors do not have to rephrase the output in their own words.

- The researcher is the passenger: **human passiv**

`Proof, Method` not mostly autonomous

Class 4
Dominant Autonomous AI Generation

- The papers's proofs and core strategy were largely written and structurally conceived by AI.
- AI was asked to prove the main statement and provided the whole proof.
- Humans or Lean has fully verified the entire proof.

`Proof, Method` mostly autonomous. It is verified!

Class 4
Dominant Autonomous AI Generation

The researcher is the passenger: **human passive everywhere**

Class 3:
the researcher chooses the intermediate steps of the problem
Class 4:
the progress of understanding was also structured by the AI.

`Proof, Method` mostly autonomous. It is verified!

Class 5
unchecked AI-generated Mathematics (OpenAI)

Mathematics generated by an LLM that was not digested, structured or understood by a human. **It was neither checked by a human nor by a formal verification tool.** This is not allowed to be put on arXiv.

Not verified!

My suggestion: The human who **curates** and understands it can add their own name to the paper and upload as class 4. The generator of the AI slop is not allowed to forbid anyone from curating and claiming the new paper.

`Class 0` — Zero AI
`Class 1` — Human-conceived discovery, AI as research assistant
`Class 2` — AI-prompted discovery with human understanding
`Class 3` — Partial AI Generation
`Class 4` — Dominant Autonomous AI Generation
`Class 5` — unchecked AI-generated Mathematics (OpenAI)

I predict that the vast majority of papers will have class 1,2 or 3.

Class 0 is very restrictive. AI can't write exposition or write code.

Class 4: AI was asked to prove the main result and it did so.

Question: Are the differences between class 1, 2 and 3 sensible?

Class 0 — Zero AI

Improving English writing

Class 1 — Human-conceived discovery, AI as research assistant

Math conversations with AI but no original ideas

Class 2 — AI-prompted discovery with human understanding

Original localized AI-ideas

Class 3 — Partial AI Generation

Entire sections originate from AI

Class 4 — Dominant Autonomous AI Generation

Whole paper originates from AI

Class 5 — unchecked AI-generated Mathematics (OpenAI)

`Class 0` — Zero AI

`Text, Lit`

`Class 1` — Human-conceived discovery, AI as research assistant

`Text, Lit, Code, Data, Idea`

`Class 2` — AI-prompted discovery with human understanding

`Check, Result`

`Class 3` — Partial AI Generation

`Class 4` — Dominant Autonomous AI Generation

`Proof, Method`

`Class 5` — unchecked AI-generated Mathematics (OpenAI)

Class 0 — Zero AI

Text, Lit

Class 1 — Human-conceived discovery, AI as research assistant

Math conversations with AI but no original ideas

Text, Lit, Code, Data, Idea

Class 2 — AI-prompted discovery with human understanding

Original localized AI-ideas

Check, Result

Class 3 — Partial AI Generation

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Class 4 — Dominant Autonomous AI Generation

Whole paper originates AI

Proof, Method

Class 5 — unchecked AI-generated Mathematics (OpenAI)

Distinguishing the `classes`

`Check`: Difference between C1 and C2

If a verification attempt falsifies the arguments or statements but suggests changes, this can lead to ideas that were first conceived by AI and not humans. This circumvents human discovery.

Distinguishing the `classes`

`Resulat`: Difference between C1, C2 (and C3)

When AI suggests changes to the assumptions, strength, or generality of a statement that do not originate in human understanding, then the AI dictates the path forward.

Distinguishing the `classes`

`Method`: Difference between C3 and C4

For mathematics that does not involve lots of compute, the label `Method` boils down to the **big picture strategy** to tackle a problem.

If this big picture strategy is proposed by AI, humans are no longer in control of the discovery process.

Distinguishing the classes

`Argument`: two separate issues
First: Difference between C1 and C2

Any argument that was not first conceived by a human. This can be very small and insignificant. Nevertheless, it detracts from the human discovery process and cedes some of the design power of further exploration to AI.

Second: Difference between C2 and C3

If the arguments, their structure, and their integration into known theory are so AI-centric that a human is simply a passenger and no longer takes part in the discovery of understanding.

Contagion Rules (important)

1. No paper of class 5 is allowed to be referenced without becoming class 5 itself and disappearing into the void of misunderstanding.

This only applies if the content of the class 5 paper is used, not if it is merely mentioned.

Contagion Rules (important)

2. A class 5 paper can be changed into a class 2-4 paper by understanding, verifying, rewriting, generalizing, specializing, explaining, presenting, etc., the contents of a class 5 paper. This process needs to be carefully refereed to ensure correctness. I do not have a clear proposal on how to referee this.

Contagion Rules (important)

3. If you cite a paper of class 3 or 4, your paper automatically becomes at least class 3 as well.

Exception: If you completely rewrite the proofs in your own words, verify every intermediate step independently, and adapt or generalize the logic to fit your framework, you can instead publish your work as class 2.

If the citation is only mentioning other work and not using it, then this rule does not apply.

Contagion Rules

4. Referencing a class 2 paper does not contaminate your class 0 or 1 status. The reason is that a class 2 paper was completely understood by a human, and the authors can explain all parts of it. Therefore, you are only drawing from the living human understanding of mathematics, which is the same thing mathematicians did before LLMs became capable of outperforming some human mathematical understanding.

Contagion Rules

5. If a paper is class 2 because of a mathematical argument that is not part of the proof of a main result, then the class can be given as `Class-2(n)` where `n` is the number of distinct contributions AI provided to change its class from 1 to 2. For every AI contribution (claim, statement, proof, proof-idea, strategy, or thought-provoking information), there needs to be an explanation. The explanation contains at least the prompt & raw AI output and the reason why it is not a core part of the paper. This special case is designed for long papers with many different statements where very few non-core results contain class-2 AI usage. The paper could have been published without these additional contributions. However, the paper becomes more powerful and useful to the community by including these additional results.

Contagion Rules

6. A class 3 or 4 paper can be changed into a class 2 paper by understanding, verifying, rewriting, generalizing, specializing, explaining, presenting, etc., the contents of a class 3 / 4 paper. This process needs to be carefully refereed to ensure human understanding. I do not have a clear proposal on how to referee this. This should only happen when a researcher is looking for new insights into the material or tries to adapt it to a different situation. A clear plan and reason should be given for attempting this work. This is contrasted by the cleaning up of a class 5 paper, which should be encouraged at any time.

Contagion Rules

7. If the contagion rule forces a paper to become say class 3 but otherwise would have been class 1, then the complete classification is 3C(1). Meaning is of class 3 by contamination, but internally it is of class 1.
The same works for 5C(4) or any other combination.